Using AWS KMS

(LAB-M13-01)

Overview

In this lab, you will:

- Configure AWS KMS to create your own customer master key (CMK)
- Encrypt EBS volumes and S3 data using AWS KMS
- Verify the behavior of S3 cross-region replication on server-side encrypted objects

Objectives

After completing this lab, you will be able to:

- Configure AWS KMS
- Encrypt Amazon EBS volumes using AWS KMS
- Encrypt Amazon S3 data using AWS KMS

Task 1: Create EC2 Instance

In this task, you will deploy EC2 instance and create file and folders in the Root volume.

Step 1: Create Amazon EC2 Instance

1. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
2. Dropdown and select **US East (N. Virginia)** region.
3. Click **Launch Instance**.
    a. In the **Choose an Amazon Machine Image (AMI)** *page*:
        i. Select the **Microsoft Windows Server 2019 Base**.
        ii. Click **Select**.
    b. In the **Choose an Instance Type** *page*:
        i. Select the **t2.micro**.
        ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** *page*:

i. **Subnet**: Dropdown and select the **First Availability Zone** (1a).

ii. Select **Next: Add Storage**.

d. In the **Add Storage** *page:*

Note: Leave all the settings as default.

i. Select **Next: Add Tags**.

e. In the **Add Tags** ***page:***

i. Click on **Add Tag**:

a) **Key**: Write **Name**.

b) **Value**: Write **Windows Server**.

ii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** *page:*

i. **Assign a security group**: Select **Create a new security group**.

a) **Security group name**: Write **WindowsServer-SG**.

b) **Description**: Write **Windows Server Security Group**.

Info: **RDP Port [3389],** already added and allowed from anywhere [0.0.0.0/0].

ii. Select **Review and Launch**.

g. In the **Review Instance Launch** *page:*

i. Select **Launch**.

a) In the **Select an existing key pair or create a new key pair** *section:*

1) **First selection box:** Dropdown and select **Choose an existing key pair**.

2) **Key pair name**: Dropdown and select **My-SEC-LAB-KP**.

3) **Enable** the **Checkmark** against the ***I acknowledge that I have access ….***

Note: Leave the other settings as default.

ii. Click on **Launch Instances**.

h. Select the **View Instances**.

Step 2: Check the Windows Server Status

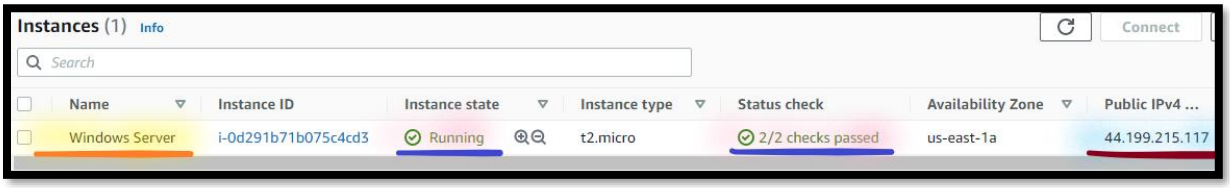
4. **From** the **EC2** console, click **Instance**.

5. Select **Windows Server**.

   a. **Wait** for the **Instance State** to change to **Running** **state**.

   b. **Wait** for the **Status check** to change to **2/2 checks passed**.

   c. **Copy** the **Public IP address** in **Notepad**.



Step 3: Generate Windows Password

6. **From** the **EC2** console, click **Instance**.

7. Select **Windows Server**.

   a. Select **Actions**.

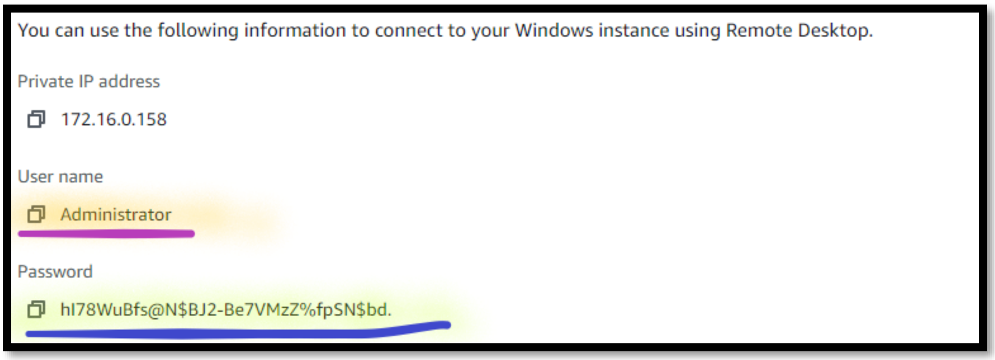
   b. Select **Security**.

c. Select **Get Windows Password**.

i. **Browse**: Navigate and select **My-SES-LAB-KP**.pem key pair.

ii. Click on **Decrypt Password**.

Note: Windows will pop-up with **user name** and **password**.



Note: **Copy** the **user name** and **password** in **Notepad**.

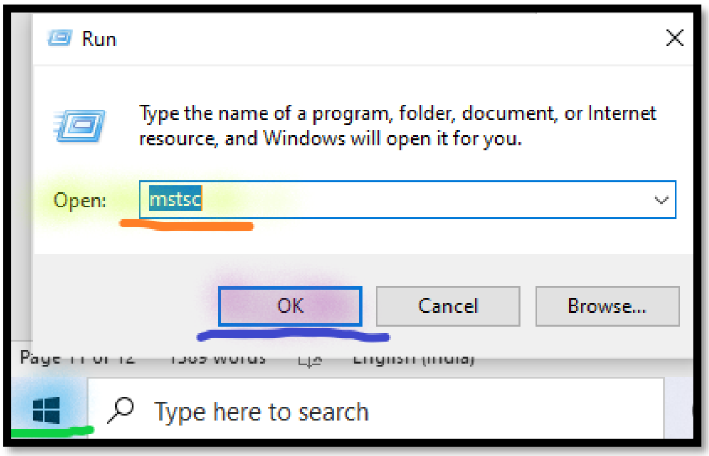
iii. Select **Close**.

Step 4: Create File and Folders in Root Volume

In this task, you will create file and folders in the Root volume.

8. **From** the **local Desktop/ Laptop**, right click on **Start** & **Run**.

a. In the open, write **mstsc**, press **Ok**.



b. **Type** the **Public IP Address** of the **Windows Server** instance.

c. Click **Connect**.

d. **Type** the **Username** and **Password** of the **Windows Server** instance.

  i. Click **Ok**.

  ii. Click on **Yes** to confirm this connection, once ***prompted with the security message.***

Create File and Folders

9. **From** the **Windows Server**:
   a. Go to **Start** menu, right click on **Start** & **Run**.
      i. In the open, write **Explorer.exe**.
      ii. Click on **This PC**.

   b. Create **Root** folder in **C drive**.

   c. Create **few** **Text** files in ***Root*** folder.

Note: **Go to next task**, But **don't logout** from **Windows server**.

Task 2: Create AWS KMS

In this task, you will create and manage an AWS KMS customer master key (CMK).

Step 1: Create AWS KMS Key

10. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

11. Dropdown and select **US East (N. Virginia)** region.

12. Select **Create a key**.

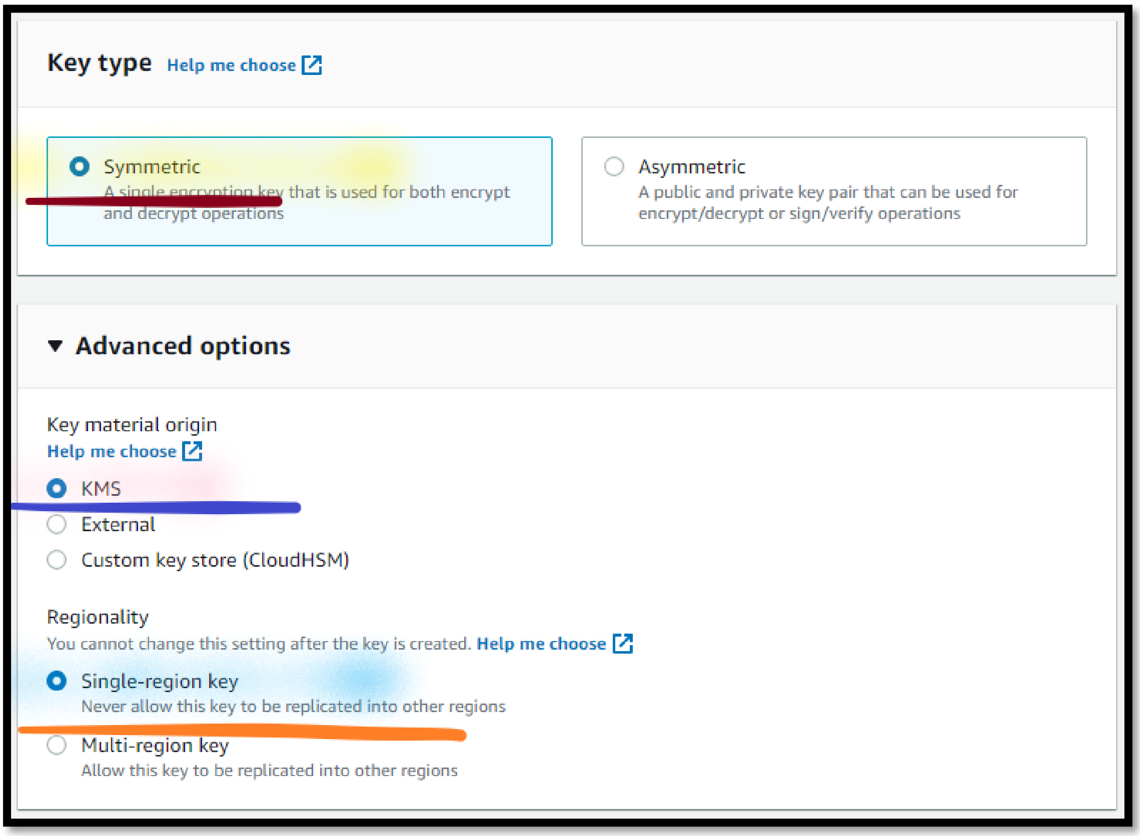
   a. In the **Configure Key** section:

      i. **Key type**: Select **Symmetric**.

      ii. **Expand** ***Advanced options***:

a) **Key material origin**: Select KMS.

b) **Regionality**: Select Single-region key.

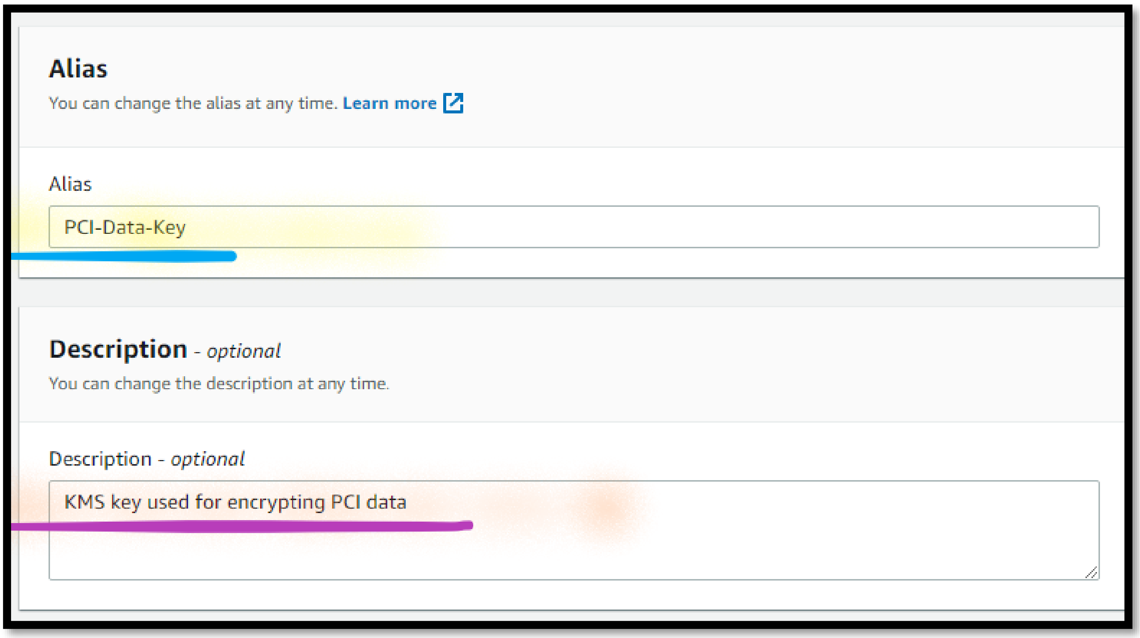


iii. Select Next.

b. In the Add labels section:

i. **Alias**: Type PCI-Data-Key.

ii. **Description**: Type KMS key used for encrypting PCI data.



iii. Click **Next**.

c. In the **Define key administrative permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

d. In the **Define key usage permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

e. In the **Review** section:

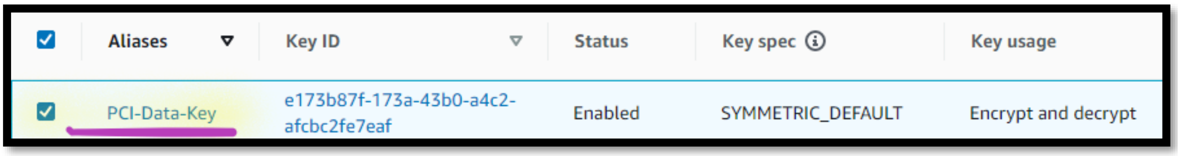
Note: Review all the details.

i. Click **Finish**.

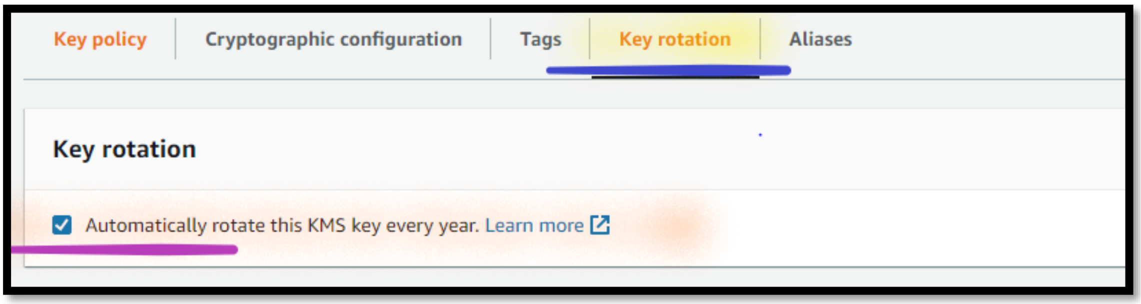
Note: Wait, till KMS key gets created.

Step 2: Rotate AWS KMS Key

13. **From** the AWS KMS Key page.
14. Select Customer managed keys.
    a. Open the PCI-Data-Key.



    b. Select the Key rotation.
        i. Enable the Checkmark against Automatically rotate this CMK every year.



        ii. Click Save.

Task 3: Encrypt Amazon EBS Volume using AWS KMS

For compliance reasons, it might be important to encrypt the storage disks of your Amazon EC2 instances. In this task, you will see how to use AWS KMS to encrypt an Amazon EBS volume and how to use AWS KMS to encrypt the boot volume of a server.

Step 1: Create Encrypted Amazon EBS volume

In this task, you will use your CMK to encrypt EBS volumes.

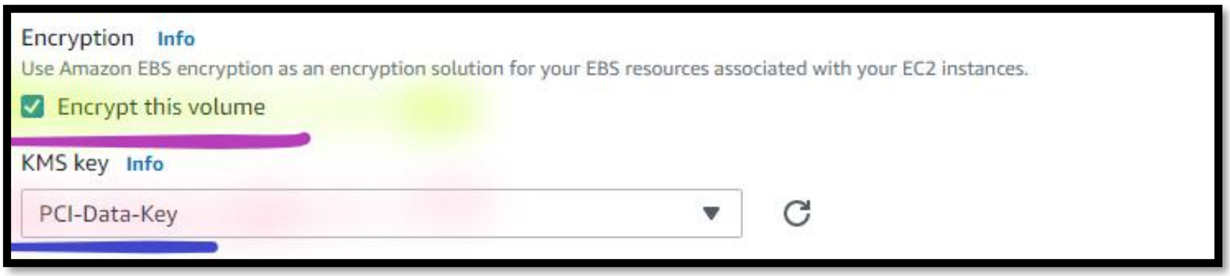
15. In the **AWS Management Console**, on the Services menu, click EC2.
16. Click Volumes.
17. Select Create volume, and configure:
    a. **Volume Type**: Dropdown and select General Purpose SSD (gp2).

b. **Size (GiB)**: Write **1**.

c. **Availability Zone**: Choose the **First Availability Zone** (1a).

d. **Encryption**: **Enable** the **Checkmark** against **Encrypt this volume**.

i. **KMS key**: Dropdown and select **PCI-Data-Key**.



Info: Encryption can't be after volume created

e. **Tag**: Select **Add tag** and **configure**:

i. **Key**: Write **Name**.

ii. **Value**: Write **WindowsServer-DataDisk**.

f. Select **Create Volume**.

Note: **Wait**, till the Volume status is showing as **available**.

Note: You will see, **Boot volume** as **Not Encrypted** and **Data Disk** as **Encrypted**.

Step 2: Attach an Encrypted Amazon EBS Volume to an Instance

18. **From** the **EC2** console, click **Volumes**.

19. Select **WindowsServer-DataDisk** volume.

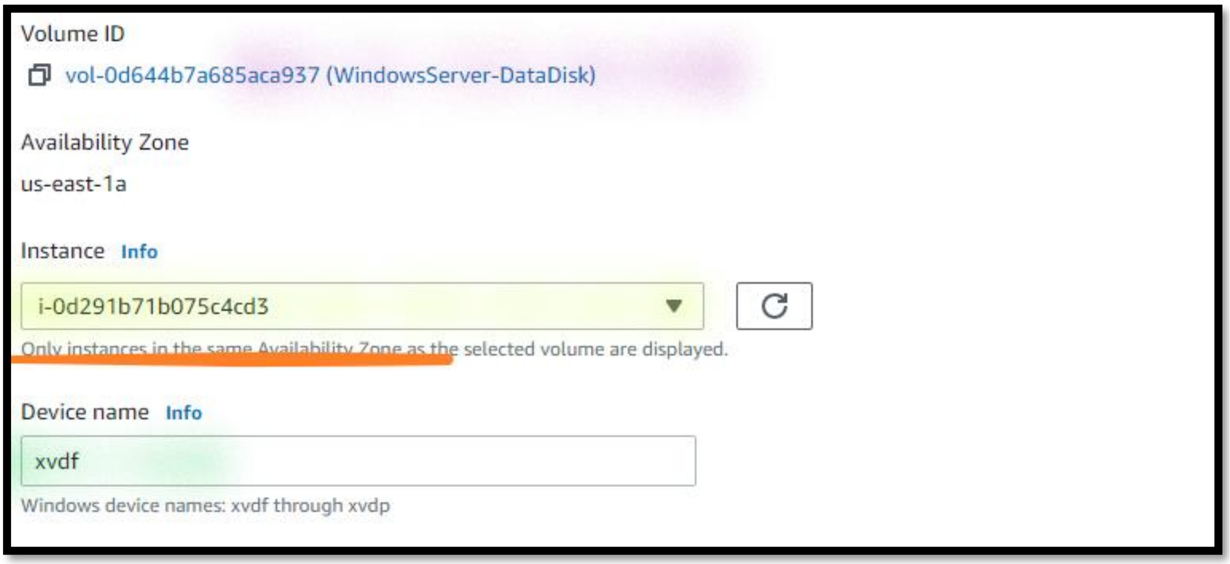
Note: Volume status is showing as **available**.

a. Choose **Actions**.

b. Select **Attach Volume**.

c. **Instance**: **Click** in the box.

i. Select **Windows Server** instance.



ii. Select **Attach volume**.

Step 3: Mount EBS Volume

In this task, you will use your encrypted EBS volume to attach to EC2 instance.

20. **Login** into **Windows Server** instance via **RDP** using **Public IP address**.

Note: Login using the **Administrator** and **Windows Password** (which you have generated in the previous step).

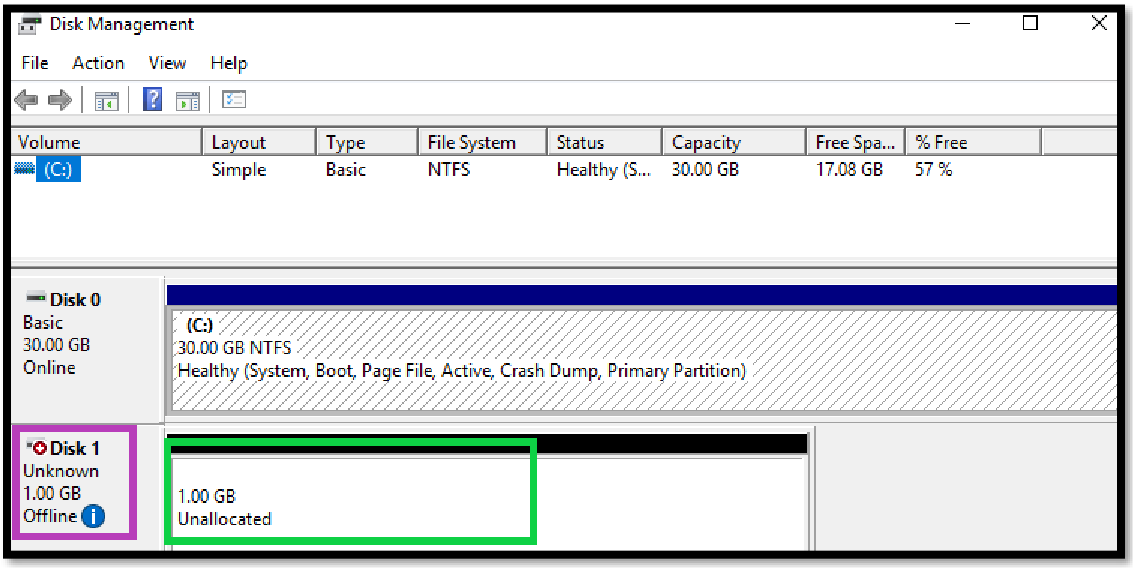
21. **From** the **Windows Server**:

    a. Go to **Start** menu, right click on **Start** & **Run**. In the open, write **diskmgmt.msc**.

    b. The new disk appears as **Unallocated**.

Note: If you are unable to see the disk, **Scroll down in Disk Management tool** to view the disk.

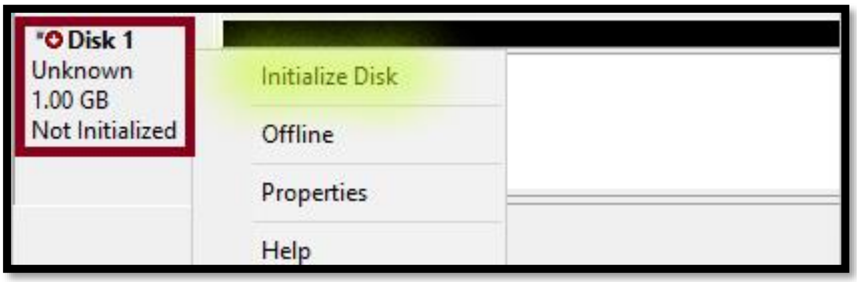
Note: Your new volume shown as **Offline**.



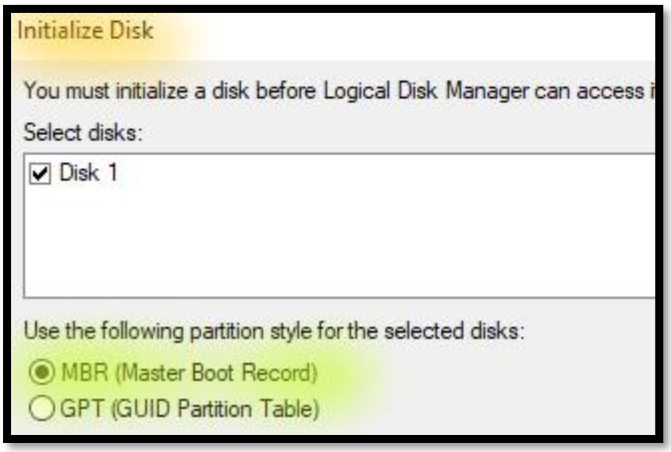
    c. **Right-click** on the disk (**within purple portion**) and select **Online**.

Note: Your new volume shown as **Not Initialised**.

d. **Right-click** on the disk (**within brown portion**) and select **Initialise Disk**.



e. **During initialise**, if it prompts you to select the partition Style, Select **MBR** (**master Boot Recorder**), select **Ok**



Note: Your new volume shown as **Online**.

f. **Right-click** on the disk (***within black portion***) and select **New simple volume**.

i. In the **Welcome to the New Simple Volume Wizard** page:

Note: Leave the other settings as default.

a) Select **Next**.

ii. In the **Specify Volume Size** page:

Note: Leave the other settings as default.

a) Select **Next**.

iii. In the **Assign Drive Letter or Path** page:

Note: Leave the other settings as default.

a) Select **Next**.

iv. In the **Format Partition** page:

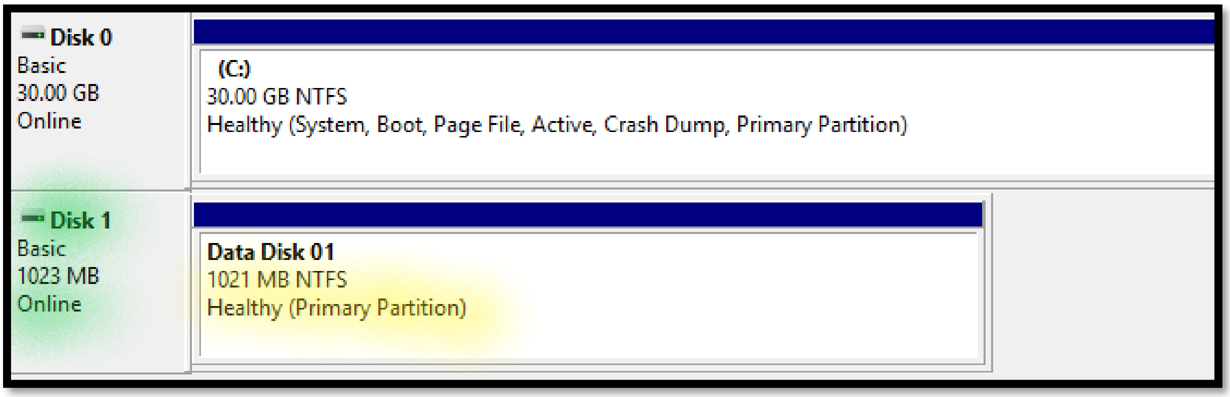
a) **Volume Label**: Write **DataDisk**.

b) Select **Next**.

v. In the **Completing the Simple Volume Wizard** page:

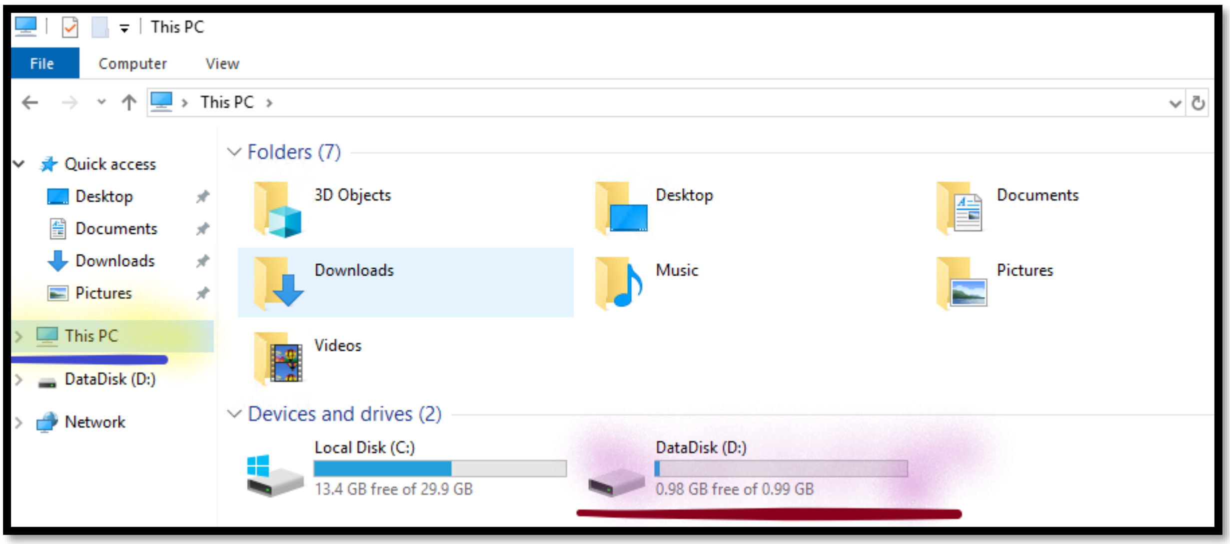
a) Select **Finish**.

Note: Once completed disk will be showing as **Healthy**. **Close disk management tool**.



Create File and Folders

22. **From** the **Windows Server**:
    a. Go to **Start** menu, right click on **Start** & **Run**.
        i. In the open, write **Explorer.exe**.
        ii. Click on **This PC**.



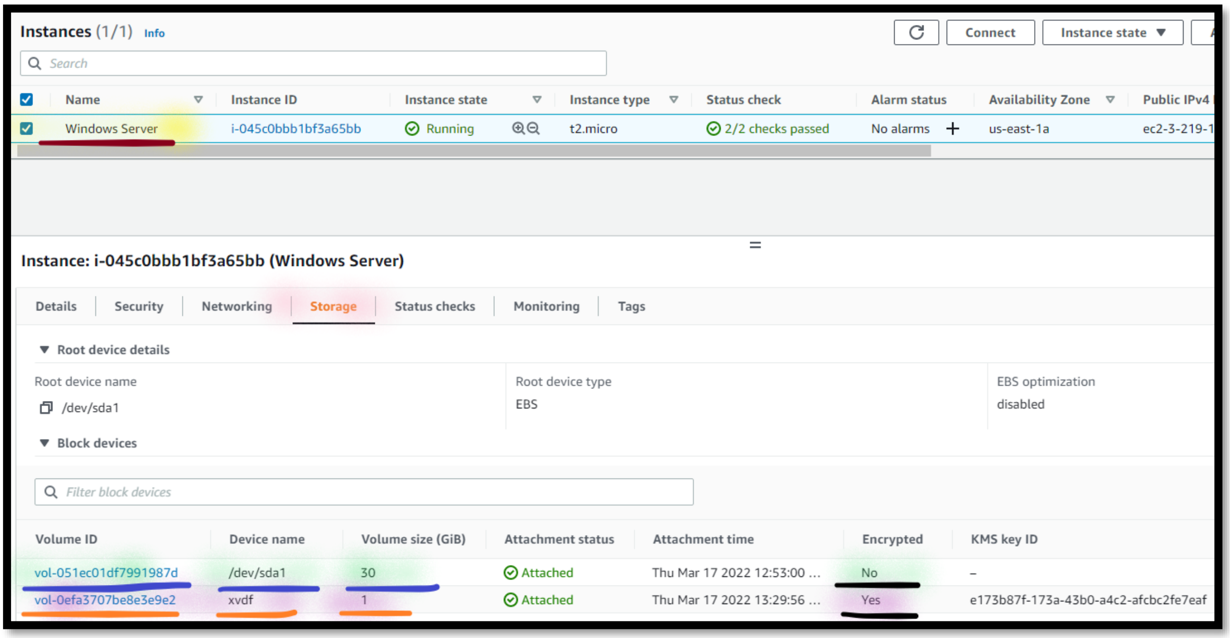
    b. Create **KMS-Disk** folder in **D drive**.
    c. Create **few** **Text** files in ***KMS-Disk*** folder.

Note: **Go to next task**, But **don't logout** from **Windows server**.

Step 4: Check the Windows Server Storage Status

23. **From** the **EC2** console, click **Instance**.
24. Select **Windows Server**.
    a. Select **Storage**.

Note: You will see, **Boot volume** as **Not Encrypted** and **Data Disk volume** as **Encrypted**.



Task 3: Encrypt a Boot volume using AWS KMS

In this section, you will use your CMK to encrypt the boot volume of an Amazon EC2 instance. This can be important when data is stored on the boot volume and you need the data to be encrypted at rest.

To encrypt a boot volume, you first need to make an Amazon Machine Image (AMI) of the instance. Then, you will encrypt the AMI by copying it. When you launch an instance from the encrypted AMI, the instance boot volume will be encrypted.

Step 1: Create Customise Image

25. **From** the **EC2** console, click **Instance**.

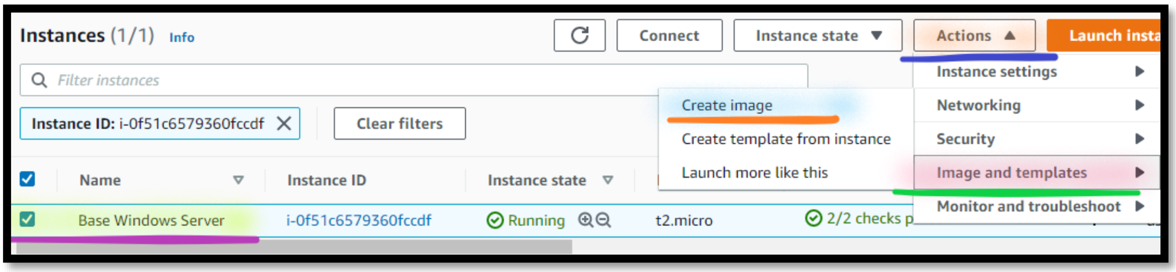
26. Select **Instances**.

    a. Select **Windows Server**.

        i. Select **Actions**.

        ii. Select **Image and templates**.

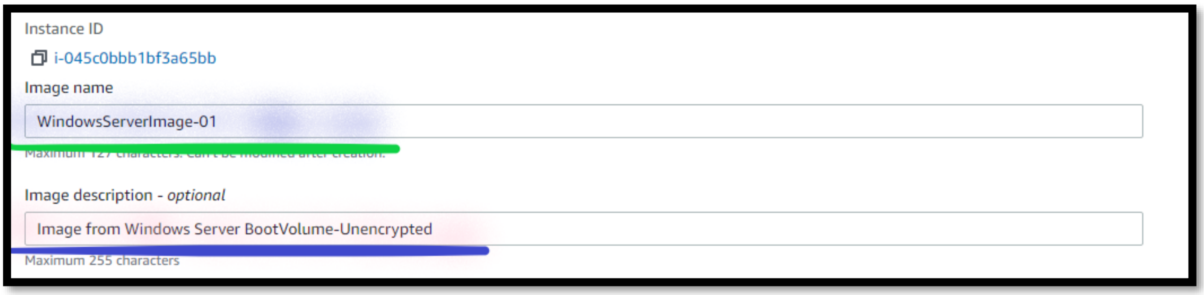
        iii. Select **Create Image**.



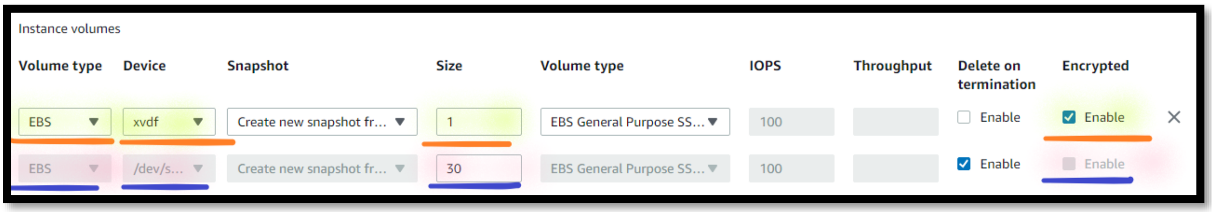
iv. In the **Create image** page:

a) **Image name**: Write **WindowsServerImage-01**.

b) **Image description**: Write **Image from Windows Server BootVolume-Unencrypted**.



Note: You will see , **Boot volume** as **Not Encrypted** and **Data Disk** as **Encrypted**.



iii. **Tag**: Select **Add tag** and **configure**:

a) **Key**: Write **Name**.

b) **Value**: Write **WindowsServerImage-01**.

v. Click on **Create Image**.

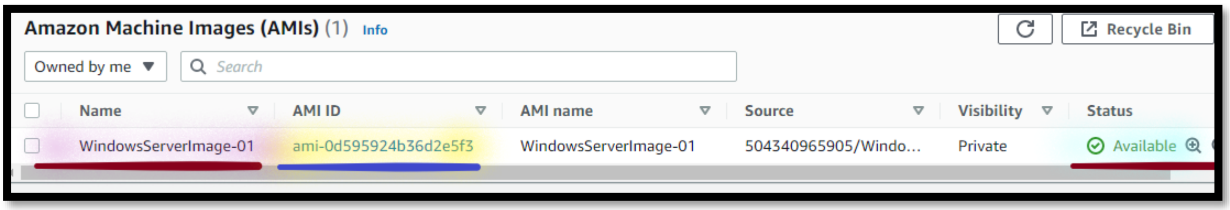
Step 2: Check the AMI and Snapshot Status

27. **From** the **EC2** console, click **AMIs**.

> **Info**: By default, create-image will restart the current instance before creating the AMI, in order to ensure the integrity of the image on the file system.

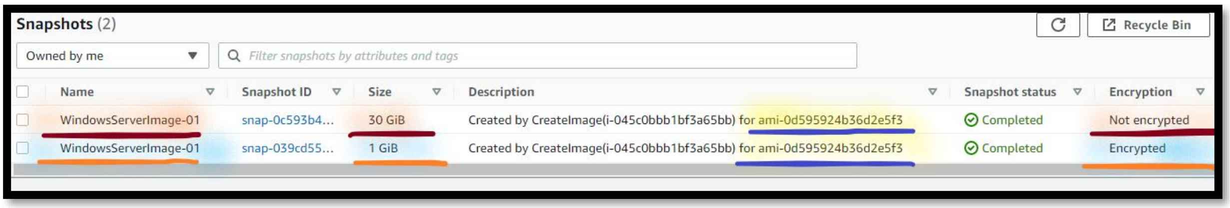
> **Note**: You will see image **Status** as **Pending**.

> **Note**: **Wait** (**~ 5 minutes**) till **Status** changed to **Available**.



28. **From** the **EC2** console, click **Snapshots**.

> **Note**: You will see, **Boot volume Snapshot** as **Not Encrypted** and **Data Disk Snapshot** as **Encrypted**.



Step 3: Create Second AWS KMS Key

29. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

30. Dropdown and select **US East (N. Virginia)** region.

31. Select **Create a key**.

    a. In the **Configure Key** section:

        i. **Key type**: Select **Symmetric**.

ii. Expand ***Advanced options***:

a) **Key material origin**: Select KMS.

b) **Regionality**: Select **Multi-region key**.

Info: AWS KMS supports multi-Region keys, which are AWS KMS keys in different AWS Regions. In multiple regions, each set of related multi-Region keys has the same key material and key ID, so you can encrypt data in one AWS Region and decrypt it in a different AWS Region without re-encrypting or making a cross-Region call to AWS KMS.

iii. Select **Next**.

b. In the **Add labels** section:

i. **Alias**: Type **PCI-Data-Key-02**.

ii. **Description**: Type **KMS key used for encrypting PCI data**.

iii. Click **Next**.

c. In the **Define key administrative permissions** section:

Note: Leave all the settings as default.

i. Click **Next**.

d. In the **Define key usage permissions** section:

Note: Leave all the settings as default.

ii. Click **Next**.

e. In the **Review** section:

Note: Review all the details.

ii. Click **Finish**.

Note: Wait, till KMS key gets created.

Step 4: Copy AMI with new KMS Key

32. **From** the **EC2** console, click **AMIs**.

a. Select **WindowsServerImage-01**.

i. Select **Actions**.

ii. Select **Copy AMI**.

a) **AMI Copy Name**: Write **WindowsServerImage-02**.

b) **Description**: Write **Image from Windows Server BootVolume_DataDisk-Unencrypted**.

c) **Destination Region**: Dropdown and select **US East (N. Virginia)** region.

d) **Enable** the **Checkmark** against **Encrypt EBS snapshots of AMI copy**.

1) **KMS key**: Dropdown and select **PCI-Data-Key-02**.

Info: Encryption keys used by an Amazon EBS volume can't be changed. However, you can Create a snapshot of the volume and then use the new encryption key.

Info: All the snapshots will be encrypted from the KMS Key, even though Volume encrypted with different KMS key.

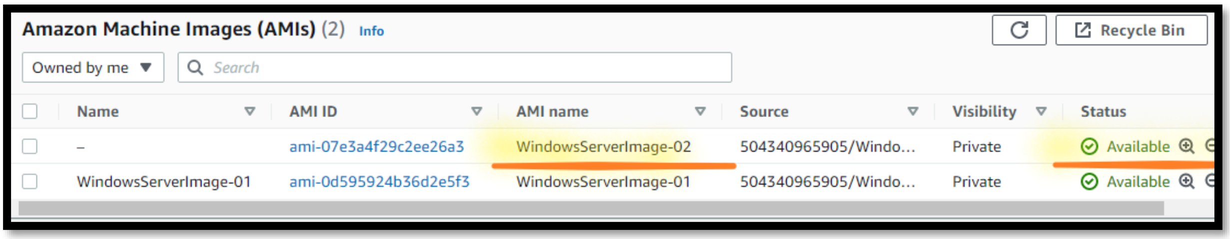
iii. Select **Copy AMI**.

Step 5: Check the AMI and Snapshot Status

33. **From** the **AMIs** page:

Note: You will see Image **Status** as **Pending**.

Note: **Wait** (~ 5 minutes) till **Status** changed to **Available**.



34. **From** the **EC2** console, click **Snapshots**.

Note: You will see, **Boot volume Snapshot** as **Encrypted** and **Data Disk Snapshot** as **Encrypted**.

Note: You will see, **Boot volume Snapshot** and **Data Disk Snapshot Encrypted** with **PCI-Data-Key-02**.

Step 6: Create Amazon EC2 Instance

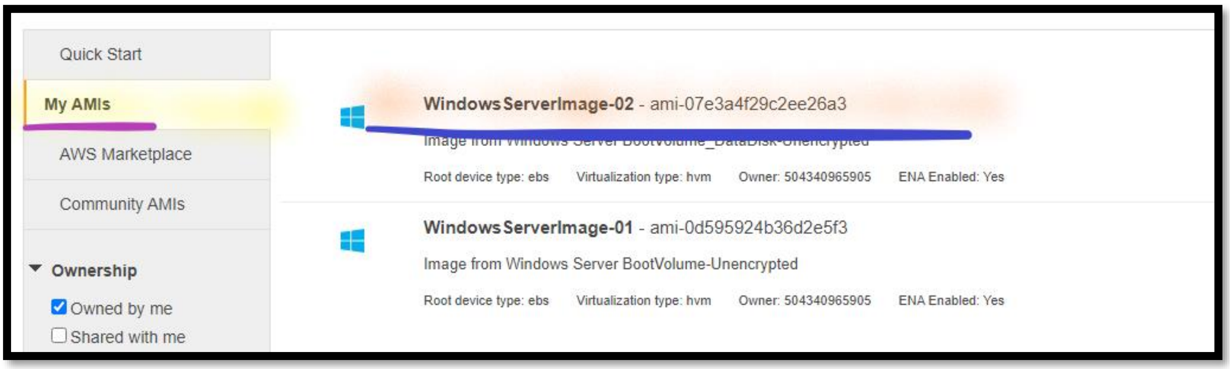
35. **From** the **EC2** console, click **Instance**.

36. Click **Launch Instance**.

   a. In the **Choose an Amazon Machine Image (AMI)** *page*:

      i. Select the **My AMIs**.

         a) Select the **WindowsServerImage-02**.



b. In the **Choose an Instance Type** *page*:

i. Select the **t2.micro**.

ii. Click **Next: Configure Instance Details**.

c. In the **Configure Instance Details** *page*:

Note: Leave all the settings as default.

i. Select **Next: Add Storage**.

d. In the **Add Storage** *page:*

Note: Leave all the settings as default.

i. Select **Next: Add Tags**.

e. In the **Add Tags** ***page:***

i. Click on **Add Tag**:

a) **Key**: Write **Name**.

b) **Value**: Write **Windows Server-02**.

ii. Select **Next: Configure Security Group**.

f. In the **Configure Security Group** *page:*

i. **Assign a security group**: Click **Select an existing security group**.

a) Select the **WindowsServer-SG**.

g. Select **Review and Launch**.

i. In the **Review Instance Launch** *page:*

a. Select **Launch**.

a) In the **Select an existing key pair or create a new key pair** *section:*

1) **First selection box:** Dropdown and select **Proceed without a key pair**.

2) **Enable** the **Checkmark** against the ***I acknowledge that I have access ….***

Note: Leave the other settings as default.

ii. Click on **Launch Instances**.

h. Select the **View Instances**.

Step 7: Check the Windows Server Status

37. **From** the **EC2** console, click **Instance**.

38. Select **Windows Server-02**.

d. **Wait** for the **Instance State** to change to **Running** **state**.

e. **Wait** for the **Status check** to change to **2/2 checks passed**.

Note: **Wait** for the server deployment.

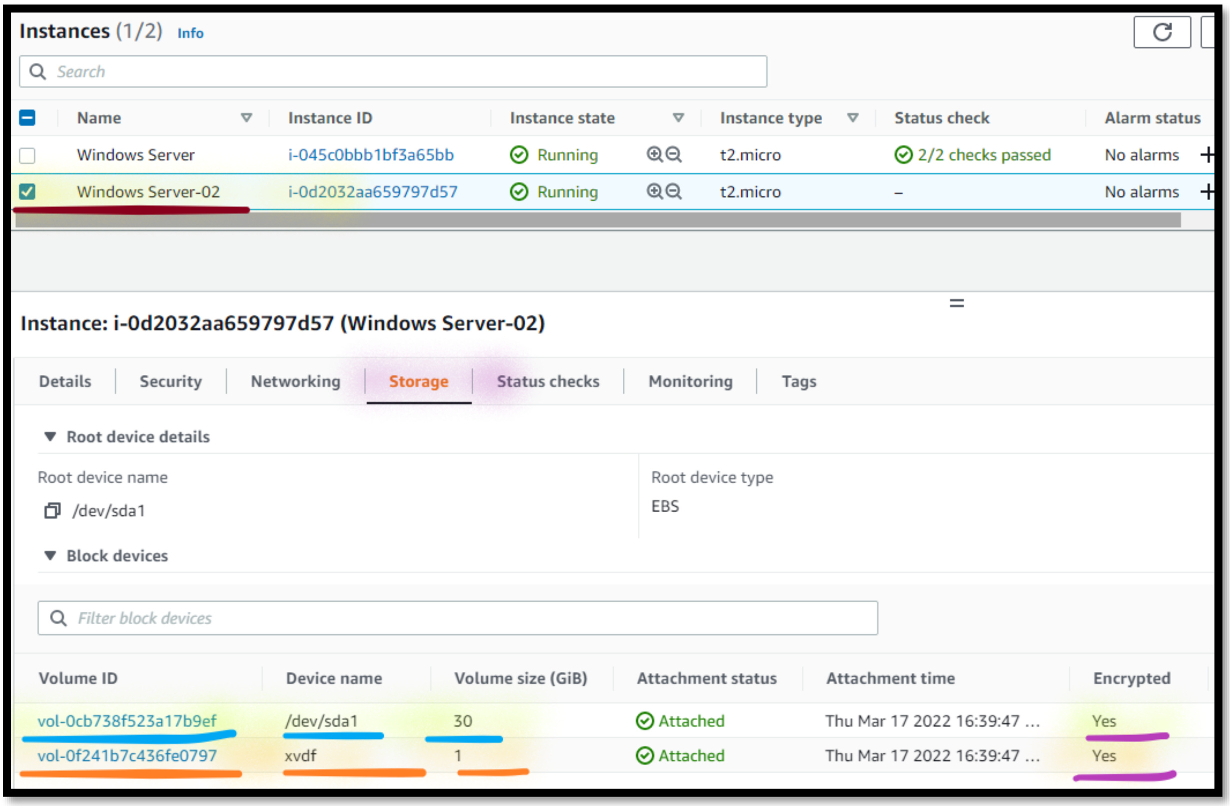
Step 8: Check the Windows Server Storage Status

39. **From** the **EC2** console, click **Instance**.

40. Select **Windows Server-02**.

a. Select **Storage**.

Note: You will see, **Boot volume** as **Encrypted** and **Data Disk volume** as **Encrypted**.



Step 10: Verify File and Folders in Second Windows Server

In this task, you will verify the file and folders created in the Root and Data Disk volume in First server.

41. **From** the **local Desktop/ Laptop**, right click on **Start** & **Run**.

   a. In the open, write **mstsc**, press **Ok**.

   b. **Type** the **Public IP Address** of the **Windows Server-02** instance.

   c. Click **Connect**.

   d. **Type** the **Username** and **Password** of the **Windows Server** (which you have generated in the First server) instance.

      i. Click **Ok**.

      ii. Click on **Yes** to confirm this connection, once ***prompted with the security message.***

42. **From** the **Windows Server-02**:
   a. Go to **Start** menu, right click on **Start** & **Run**.
      i. In the open, write **Explorer.exe**.
      ii. Click on **This PC**.
   b. **Open** the **Root** folder in **C drive**.
      i. **Open** and **verify** the **Text** files in ***Root*** folder.
   c. **Open** the **KMS-Disk** folder in **C drive**.
      i. **Open** and **verify** the **Text** files in ***Root*** folder.

Task 4: Encrypt Amazon S3 data using AWS KMS

In this task, you will encrypt Amazon S3 data using your CMK. This method of encryption is referred to as Amazon S3 server-side encryption with AWS KMS managed keys (SSE-KMS).

Step 1: Encrypt Amazon S3 data

43. In the **AWS Management Console**, on the **Services** menu, click **S3**.
44. Click **Create bucket**.
   a. In the **General configuration** section:
      i. **Bucket name**: Write **secret-123**.

Note: **Replace 123** with a random number to make bucket name unique.

      ii. **Region**: Dropdown and select **US East (N. Virginia)**.

b. In the **Object Ownership** section:

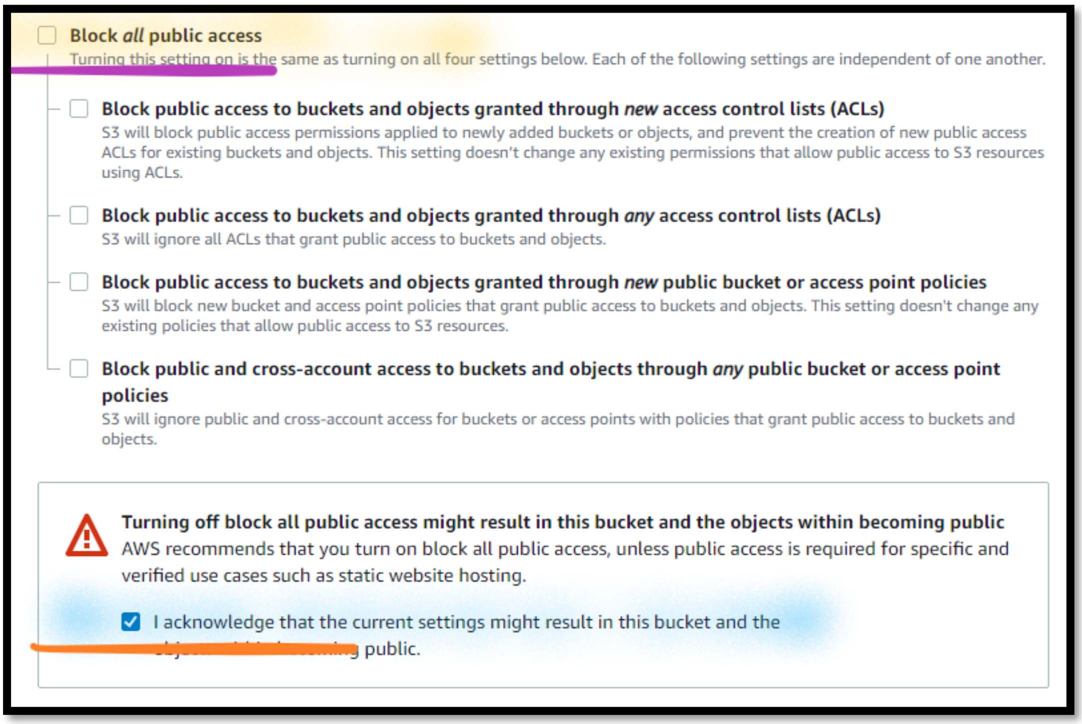
   i. Select **ACL Enabled**.

   ii. Select **Bucket owner preferred**.

c. In the **Block Public Access settings for this bucket** section:

   i. **Block all public access**: **Deselect** (***remove the checkmark***) the Block all public access option.

   ii. **Select** **I acknowledge** *that the current settings might result in this bucket and the objects within becoming public*.



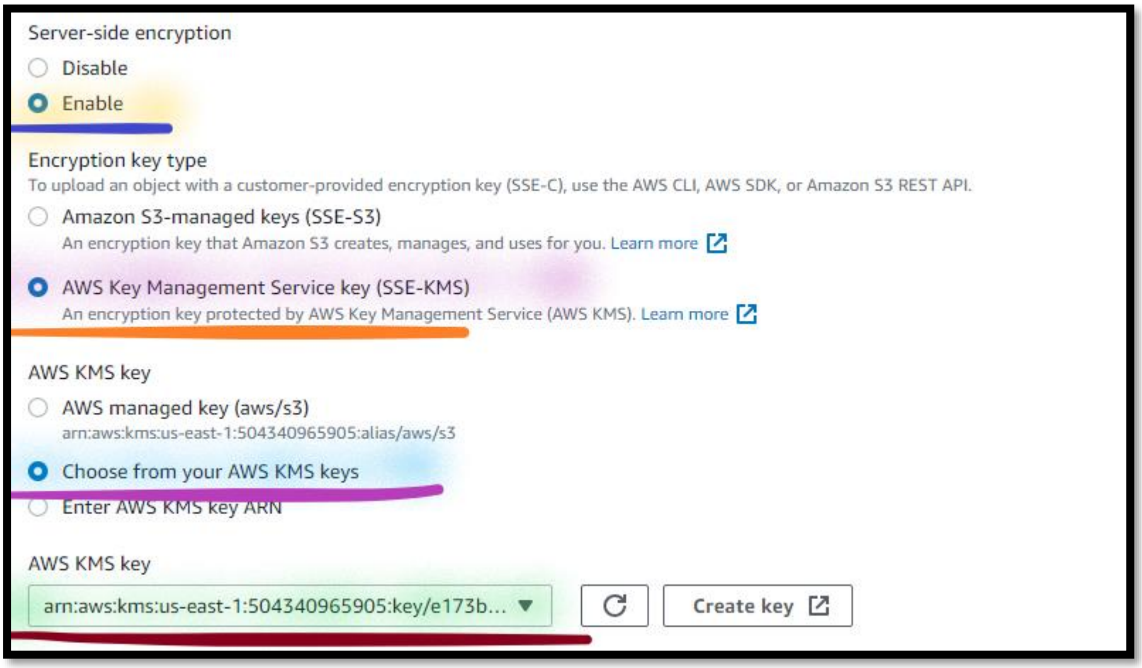
d. In the **Default encryption** section:

   i. **Server-side encryption**: Select **Enable**.

   ii. **Encryption key type**: Select **AWS Key Management Service key (SSE-KMS)**.

   iii. **AWS KMS key**: Select **Choose from your AWS KMS keys**.

   iv. **AWS KMS key**: Dropdown and select **PCI-Data-Key-02**.



Note: Leave other details as default.

e. Click **Create bucket**.

Step 2: Upload the Object in the Bucket

45. **From** the **S3** console, click **Buckets**.

46. Open **secret-123** bucket.

a. Click **Objects**.

b. Click **Upload**.

c. Select **Add files**.

d. Navigate and select **mydata.txt** file.

Note: **mydata.txt** file available with the lab manual.

e. Click **Upload**.

Note: **Wait** till all the contents uploaded succesfully and can get the message **Upload succeeded**.

f. Click **Close**.

Step 3: Make Object Public

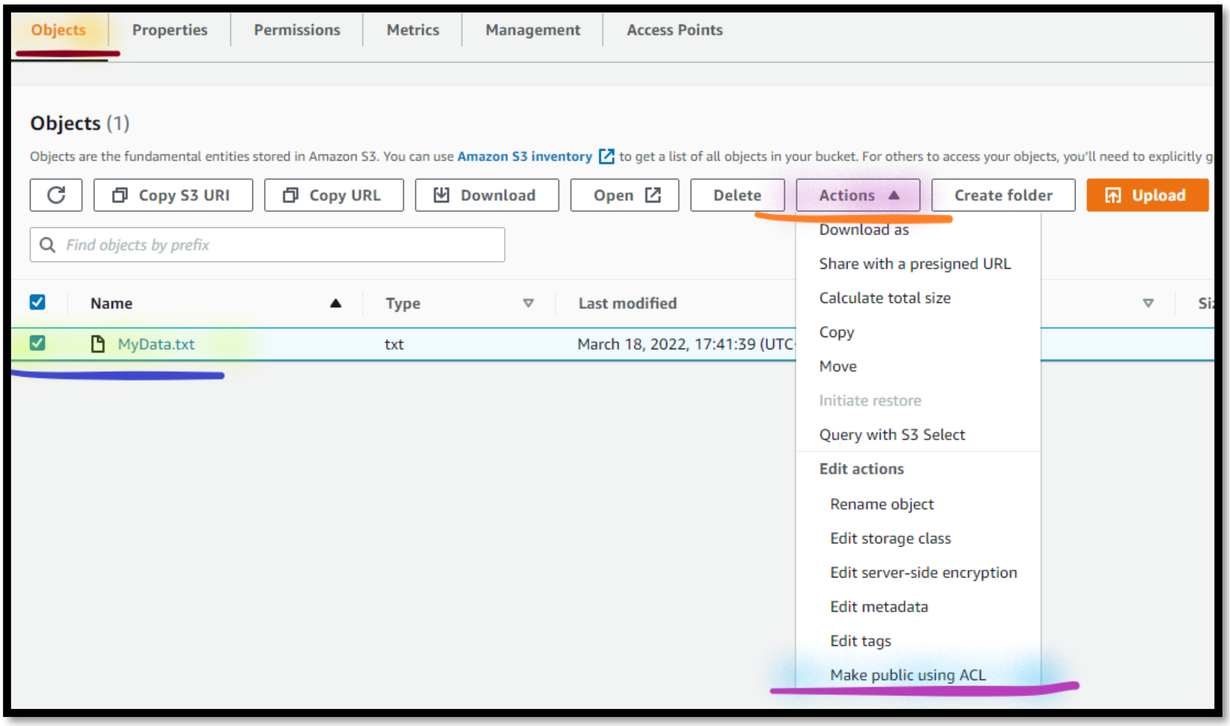
47. **From** the S3 console, click **Buckets**.

48. Open **secret-123** bucket.

   a. Select **mydata.txt**.

      i. Select **Actions**.

      ii. Select **Make public using ACL**.



a) Select **Make public**.

b) Select **Close**.

Step 4: Access the Public Object

49. **From** the S3 console, click **Buckets**.

50. Open **secret-123** bucket.

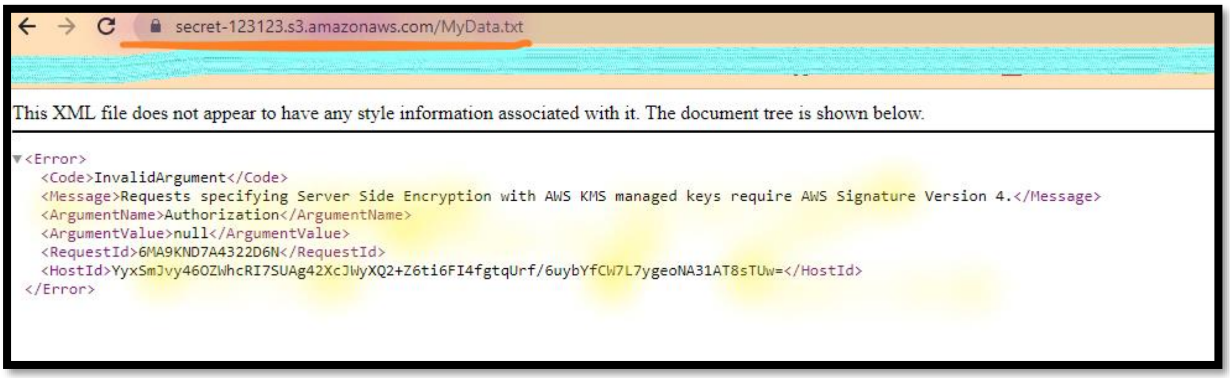
   a. Open **mydata.txt**.

   b. Copy **Object URL** in the **Notepad**.

17. **From** your **local Desktop/ Laptop** open the **Browser**, type **Object URL** of ***MyData.txt*** and access your ***Object***.

Note: You are **not able to open the file**.

Info: Notice that you are not able to open the file, even though it is public, because the file is encrypted and needs to be decrypted before you can view it. The error message also tells you that the reason the file cannot be opened is because the request requires an AWS KMS key managed by AWS.



18. **From** the **S3** console, click **Buckets**.

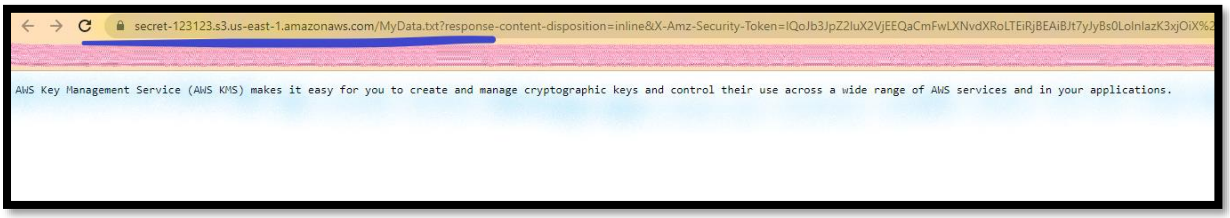
19. Open **secret-123** bucket.

    a. Open **mydata.txt**.

    b. Click the **Open**.

Note: You are **able to open the file**.

Info: You are able to see the file because Amazon S3 transparently decrypts the file using your CMK as long as the request comes from a source that has permission to use the CMK.



Note: You can also use Presigned URL to access the Encrypted Object.

Note: **Go to next task**, But **don't close** the **Browser**.

Step 5: Disable the CMK

20. In the **AWS Management Console**, on the **Services** menu, select **KMS**.
21. Select **Customer managed keys**.
    a. Open the **PCI-Data-Key**.
        i. Select **Key action**.
        ii. Select **Disable**.
            a) **Enable** the **Checkmark** against **Confirm that you want to disable this key**.
            b) Select **Disable key**.

Note: You can see **the PCI-Data-Key status** as **Disable**.

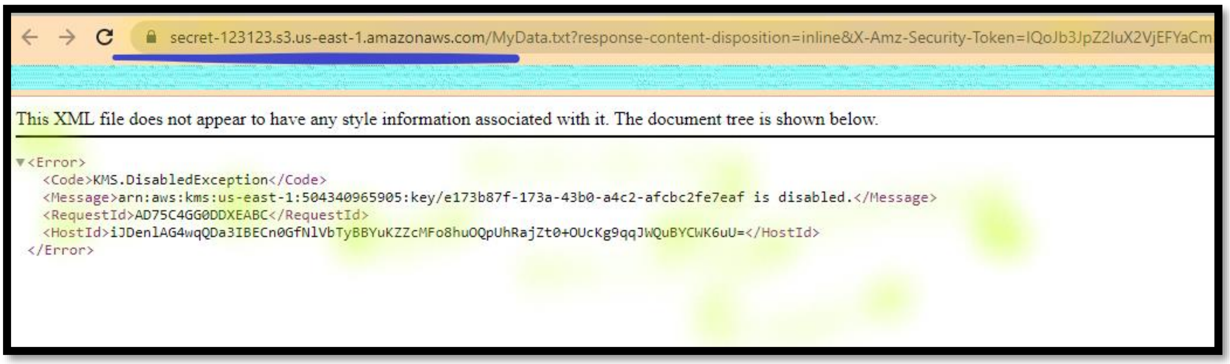
Step 6: Access the Public Object

17. **From** the **S3** console, click **Buckets**.
18. Open **secret-123** bucket.
    a. Open **mydata.txt**.
    b. Click the **Open**.

Note: You are **not able to open the file**.

Note: Notice the error code and message. You are not able to view the file because the key is disabled.

Note: If you are able to view the file, perform a hard refresh (Press CTRL+F5) in your browser as the file may be cached.



Task 5: Create IAM Role

Step 1: Create IAM Policy for AWS S3

51. In the **AWS Management Console**, on the **Services** menu Search and Select **IAM**.
52. Select **Policies**.
53. Click on **Create policy**.
    a. Select the **JSON**:
        i. **Remove** the ***existing policy***.

ii. **Copy** the policy from **S3ReplicationPolicy**.txt.

Note: **S3ReplicationPolicy**.txt is available with the Lab manual.

iii. Select **Next: Tags**.

iv. Select **Next: Review**.

b. In the **Review Policy** page:

i. **Name**: Write **S3ReplicationPolicy**.

ii. Click on **Create Policy**.

Note: You will get the message **S3ReplicationPolicy** has been created.

Step 2: Create IAM Roles for AWS S3

54. **From** the **IAM** console:

55. Select **Roles**, click on **Create role**.

a. In the **Select trusted entity** section:

i. **Trusted entity type**: Select **AWS service**.

ii. **Common use cases**: Dropdown and select **S3**.

a) Select **S3**.

iii. Select **Next**.

b. In the **Add Permissions** section:

i. in the **Search box**, type **S3ReplicationPolicy**, press **Enter**.

ii. Select **S3ReplicationPolicy**.

iii. Select **Next**.

c. In the **Name, review, and create** section:

i. **Role Name**: Write **S3ReplicationRole**.

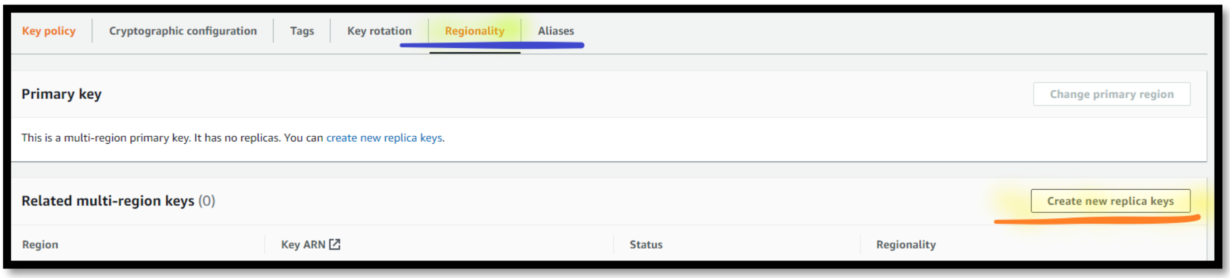
ii. Click **Create role**.

Note: **Wait**, till role **S3ReplicationRole** gets created.

Task 6: AWS S3 Cross-region Replication

Step 1: Create AWS KMS Replica

56. In the **AWS Management Console**, on the **Services** menu, select **KMS**.
57. Select **Customer managed keys**.
    a. Open the **PCI-Data-Key-02**.
        i. Select the **Regionality**.
        ii. Select **Create new replica keys**.

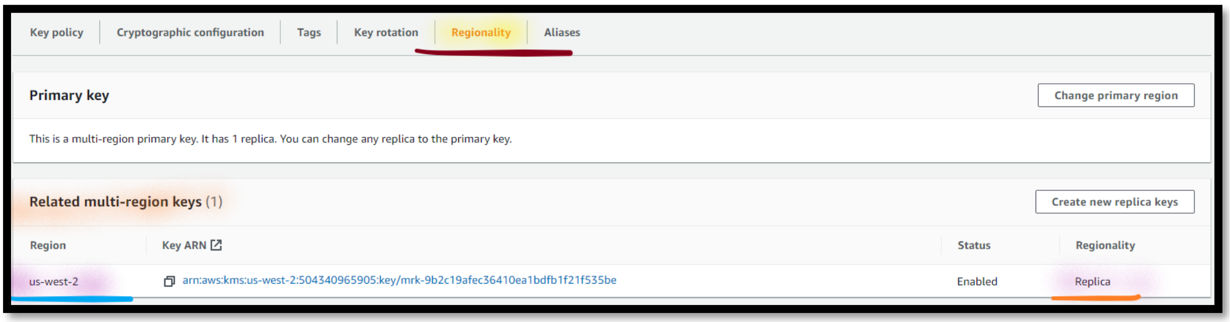


a) **Choose regions for your new replica key**: Dropdown and select **US West (Oregon)** [us-west-2].
b) Select **Next**.
c) Select **Next**.
d) Select **Next**.
e) Select **Next**.
f) Select **Create new replica keys**.
g) **Enable** the **Checkmark** against **I understand that the values I choose here are not synchronised with any other multi-region key**.

58. **From** the **KMS** console:
    a. Select **Customer managed keys**.
        i. Open the **PCI-Data-Key-02**.

a) Select the **Regionality**.

Note: You can see the **KMS Replica** details of **us-west-2**.



Step 2: Verify AWS KMS Replica

59. Dropdown and select **US West (Oregon)** region.

60. In the **AWS Management Console**, on the **Services** menu, select **KMS**.

    a. Select **Customer managed keys**.

Note: You can see the **PCI-Data-Key-02** details.

Note: Verify **Key ID** is same in **Primary** and **Replica** region.

Step 3: Enable Versioning for AWS S3 Bucket

61. In the **AWS Management Console**, on the **Services** menu, select **S3**.

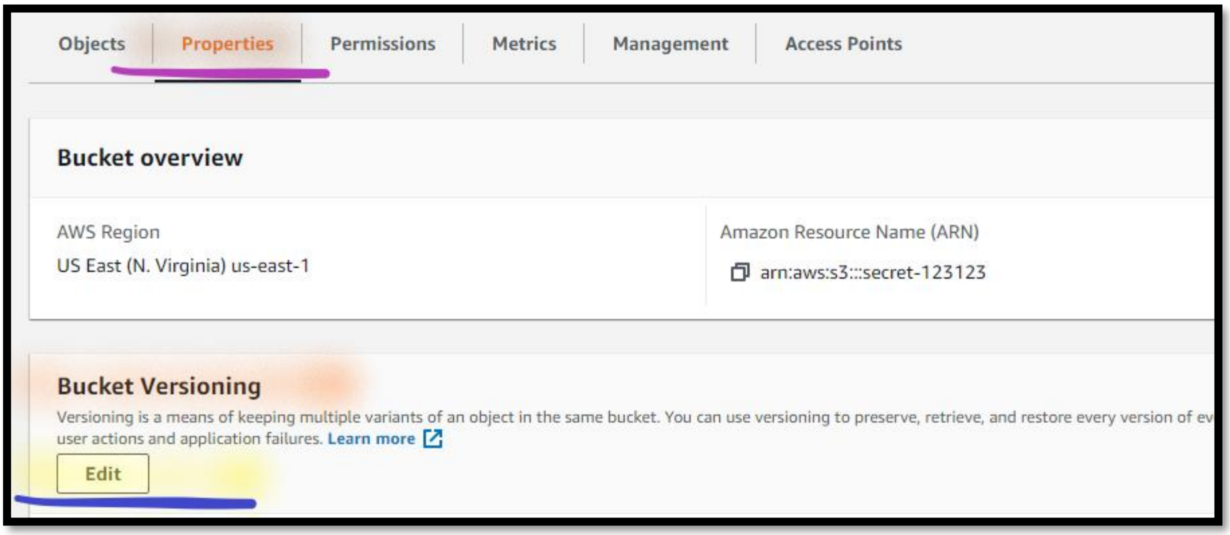
62. Select **Buckets**.

    a. Open **secret-123** bucket.

    b. Select **properties**.

    c. In the **Bucket Versioning** section:

        i. Select **Edit**.



ii. **Bucket Versioning**: Select **Enable**.

iii. Select **Save Changes**.

Step 4: Create Second AWS S3 Bucket

63. **From** S3 console:

64. Click **Create bucket**.

a. In the **General configuration** section:

i. **Bucket name**: Write **replica-123**.

ii. **Region**: Dropdown and select **US West (Oregon)**.

b. In the **Object Ownership** section:

i. Select **ACL Enabled**.

ii. Select **Bucket owner preferred**.

c. In the **Block Public Access settings for this bucket** section:

i. **Block all public access**: **Deselect** (***remove the checkmark***) the Block all public access option.

ii. **Select** **I acknowledge** *that the current settings might result in this bucket and the objects within becoming public*.

d. In the **Bucket Versioning** section:

i. **Bucket Versioning**: Select **Enable**.

Note: Leave other details as default.

e. Click **Create bucket**.

Step 5: Enable Cross-region Replication

65. **From** **S3** console:

66. Select **Buckets**.

a. Open **secret-123** bucket.

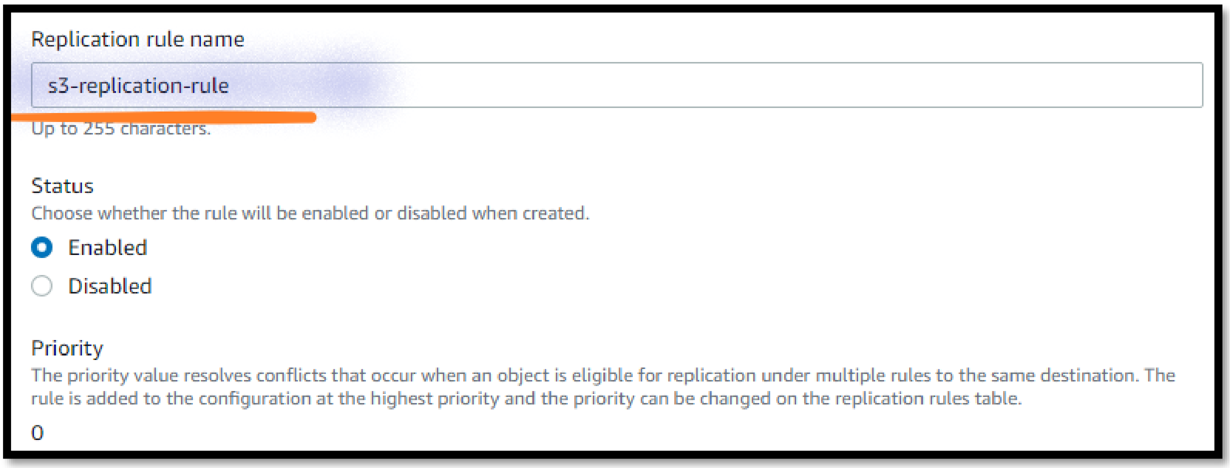
b. Select **Management**.

c. In the **Replication rules** section:

i. Select **Create replication rule**:

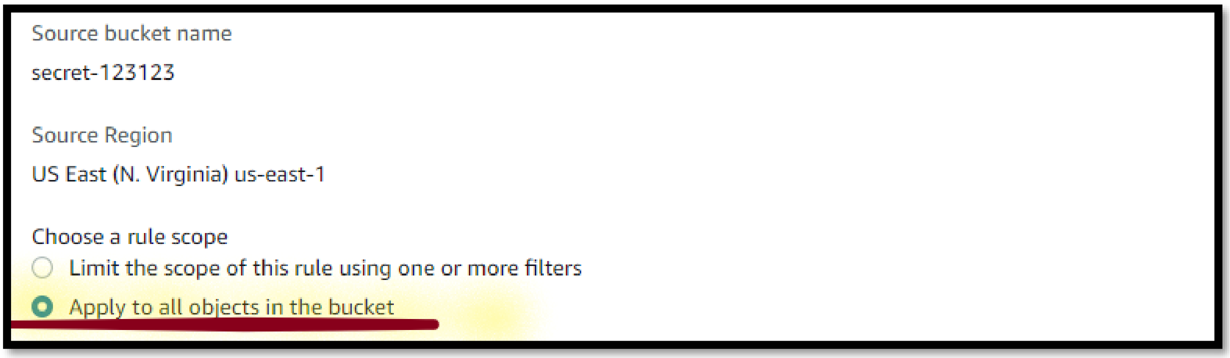
a) In the **Replication rule configuration** section:

1) **Replication rule name**: Write **s3-replication-rule**.



b) In the **Source bucket** section:

1) **Choose a rule scope**: Select **Apply to all objects in the bucket**.



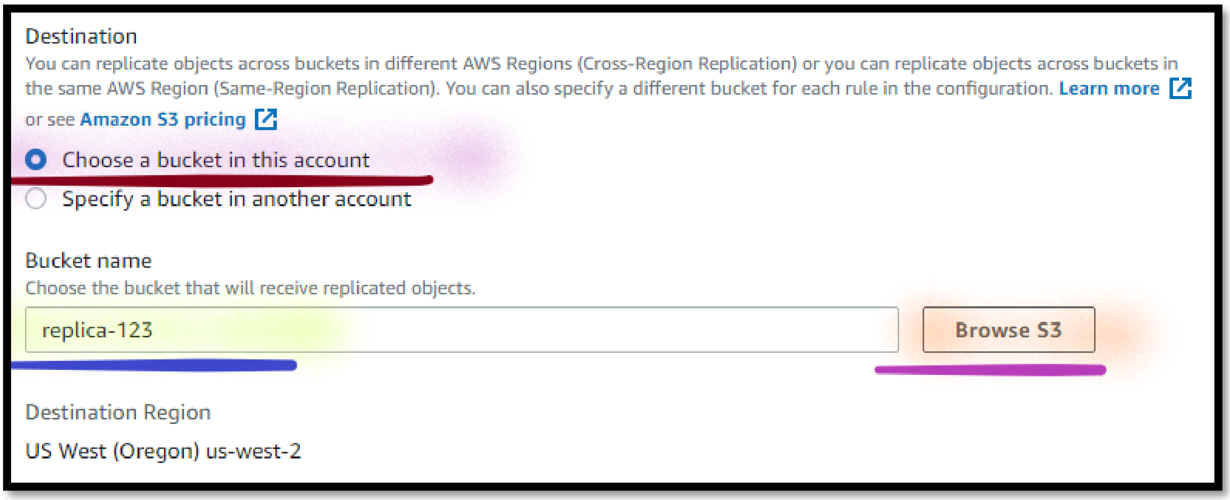
c) In the **Destination** section:

1) Select **Choose a bucket in this account**.

2) **Bucket name**: Select **Browse**:

I. Select the **replica-123** bucket.

II. Select **Choose path**.



III. **Choose a rule scope**: Select **Apply to all objects in the bucket**.

d) In the **IAM role** section:

1) Select **Choose from existing IAM roles**.

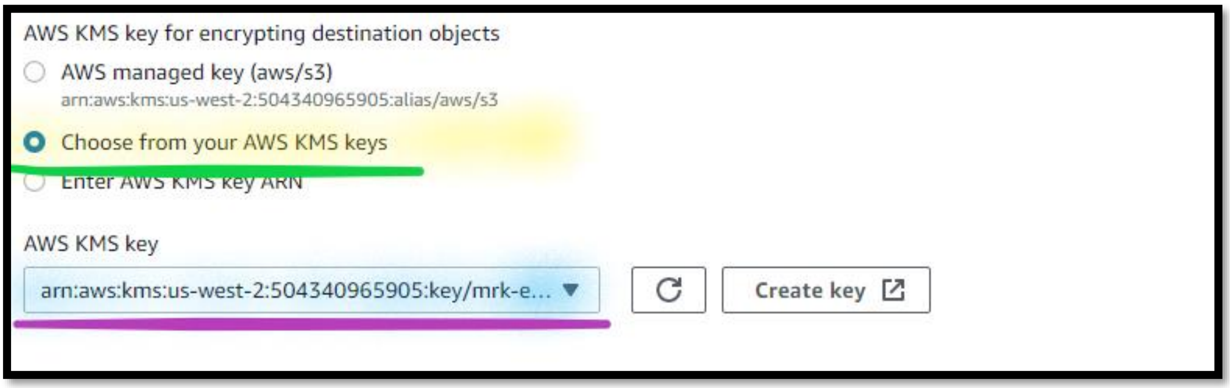
2) **IAM role:** Dropdown and select **S3ReplicationRole**.

e) In the **Encryption** section:

1) Select **Replicate objects encrypted with AWS KMS**.

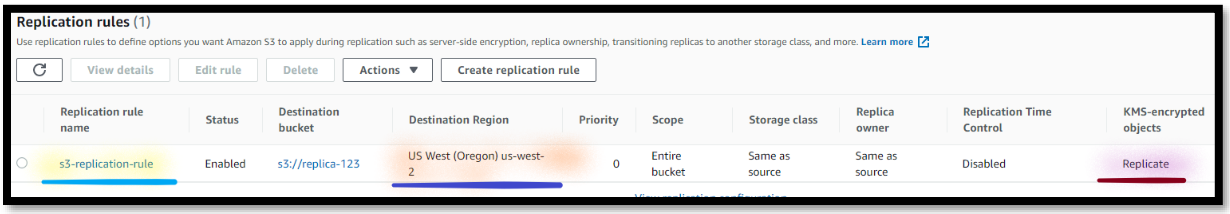
2) **AWS KMS key for encrypting destination objects**: Select **Choose from your AWS KMS keys**.

3) **AWS KMS key**: Dropdown and select **PCI-Data-Key-02**.



f) Select **Save**.

Note: You can see the **Replication rules**.



Step 6: Upload the Object in the Bucket

67. **From** the **S3** console, click **Buckets**.

68. Open **secret-123** bucket.

a. Click **Objects**.

b. Click **Upload**.

c. Select **Add files**.

d. Navigate and select **myreplica.txt** file.

Note: **myreplica.txt** file available with the lab manual.

e. Click **Upload**.

Note: **Wait** till all the contents uploaded succesfully and can get the message **Upload succeeded**.

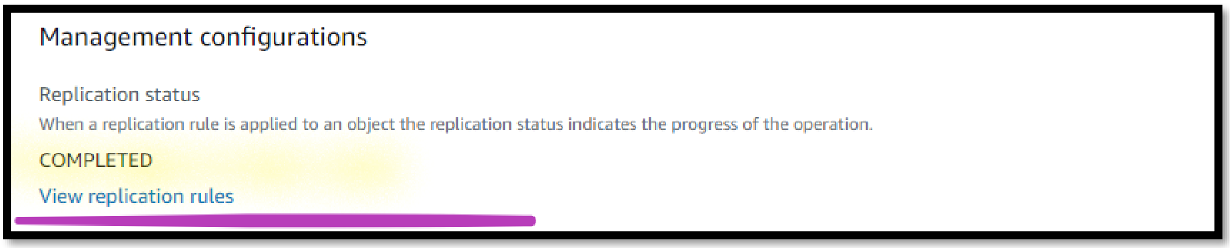
f. Click **Close**.

Step 7: View the Object status

69.**From** the **Replica-123** bucket console:

a. Open the **myreplica.txt** file.

Note: **Wait** till the **Replication Status** is **Completed**.

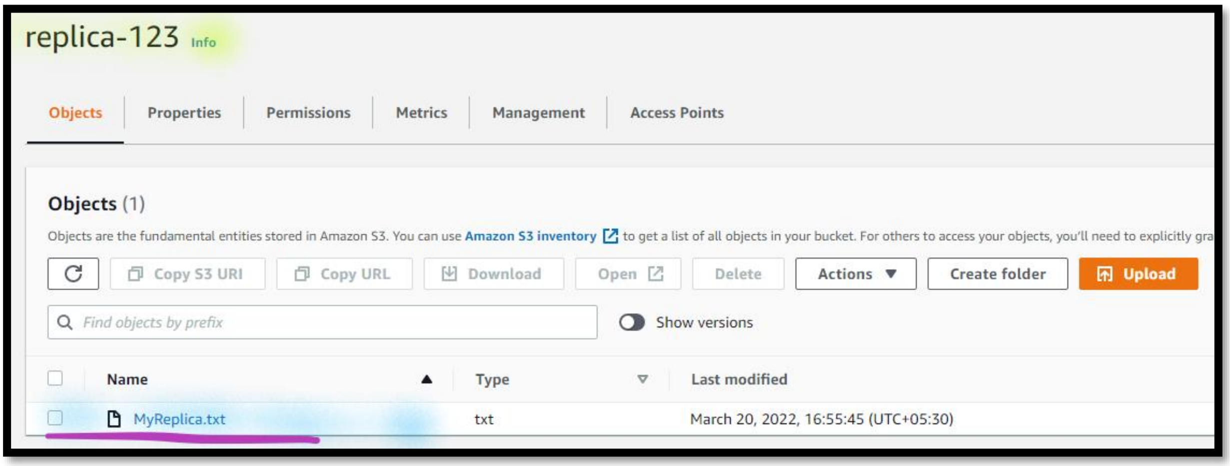


Step 8: Access the Replicated Object

70.**From** the **S3** console, click **Buckets**.

71.Open **replica-123** bucket.

Note: You can see the **myreplica.txt** file.



a. Open **myreplica.txt** file.

b. Click the **Open**.

Note: You are **able to open the file**.

Task 4: Cleanup the Environment

Step 1: Terminate EC2 Instances

72. In the **AWS Management Console**, on the **Services** menu, click **EC2**.
73. Click **Instances**.
    a. Select the **Two** instances.
        i. Click on **Instance state**.
        ii. Select **Terminate instance**.
        iii. Select **Terminate**.

Step 2: Delete AMIs

74. **From** the **EC2** console, click **AMIs**.
    a. Select **WindowsServerImage-01**.
        i. Select **Actions**.
        ii. Select **Deregister AMI**.
        iii. Select **Deregister AMI**.
    b. Select **WindowsServerImage-02**.
        i. Select **Actions**.
        ii. Select **Deregister AMI**.
        iii. Select **Deregister AMI**.

Step 3: Delete Snapshots

75. **From** the **EC2** console, click **Snapshots**.
    a. Select **All** the snapshots.

i. Select **Actions**.

ii. Select **Delete Snapshot**.

a) In the **prompt**, type **delete**.

b) Select **Delete**.

Step 4: Delete Volumes

76. **From** the **EC2** console, click **Volumes**.

a. Select **All** the volumes.

i. Select **Actions**.

ii. Select **Delete volume**.

a) In the **prompt**, type **delete**.

b) Select **Delete**.

Step 5: Delete the S3 Bucket

77. In the **AWS Management Console**, on the **Services** menu, click **S3**.

78. Select **Buckets**.

Delete the First bucket

79. Select **secret-123** bucket.

a. Select **Empty**.

b. **Type** **permanently delete** to ***Delete all the objects*.**

c. Select **Empty**.

d. Select **Exit**.

80. Select **secret-123** bucket.

a. Select **Delete**.

b. **Type** **secret-123** bucket name, to ***Delete bucket***.

c. Select **Delete bucket**.

Delete the Second bucket

81. Select **replica-123** bucket.
    a. Select **Empty**.
    b. **Type** **permanently delete** to ***Delete all the objects*.**
    c. Select **Empty**.
    d. Select **Exit**.
82. Select **replica-123** bucket.
    a. Select **Delete**.
    b. **Type** **replica-123** bucket name, to ***Delete bucket***.
    c. Select **Delete bucket**.

Step 6: Schedule the CMK for deletion

83. In the **AWS Management Console**, on the **Services** menu Search and Select **KMS**.
84. Dropdown and select **US East (N. Virginia)** region.
85. Select **Customer managed keys**.

Delete the First CMK

a. Open the **PCI-Data-Key**.
   i. Select **Key actions**.
   ii. Select **Schedule key deletion**.
   iii. **Waiting period (days)**: Type **7**.
   iv. **Select** ***Confirm that you want to schedule these keys for deletion after a 7 day waiting period***.
   v. Select **Schedule deletion**.

Delete the Second CMK

b. Open the **PCI-Data-Key-02**.
   i. Select **Key actions**.
   ii. Select **Schedule key deletion**.

iii. **Waiting period (days)**: Type **7**.

iv. **Select** ***Confirm that you want to schedule these keys for deletion after a 7 day waiting period***.

v. Select **Schedule deletion**.

Delete the Replica CMK

86. **From** the **KMS** console:

87. Dropdown and select **US West (Oregon)** region.

a. Open the **PCI-Data-Key-02**.

i. Select **Key actions**.

ii. Select **Schedule key deletion**.

iii. **Waiting period (days)**: Type **7**.

iv. **Select** ***Confirm that you want to schedule these keys for deletion after a 7 day waiting period***.